Data Analysis and EDA Report

1. Objective of This Analysis

The purpose of this analysis is to:

- Understand the structure and characteristics of the dataset
- Identify data quality issues
- Provide insights that guide data cleaning, feature engineering, and model building

This document focuses **only on data understanding and exploration**, not modeling.

2. Dataset Overview

The dataset consists of multiple relational tables centered around **patient ICU stays**, including:

- Patients (demographics)
- Admissions
- Diagnoses
- Medications
- Laboratory results
- Vital signs (periodic)

The common linking key across tables is:

- `patientunitstayid`

Each table represents **one-to-many relationships** with patients.

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3. Patients Table – Key Findings

Key variables analyzed

- **patientunitstayid**
- **gender**
- **ethnicity**
- **admissionweight**
- **unittype**
- **apacheadmissiondx**
- **hospitaldischargestatus**

Findings

- Gender distribution is imbalanced (dominant categories observed).

- Ethnicity contains multiple categories with long-tail distributions.

Implications

- Demographic imbalance should be considered during model evaluation.
- Ethnicity may require grouping or encoding strategies.

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4. Admissions Table – Key Findings

Key variables

- Admission diagnosis names and paths
- Diagnosis entry offsets

Findings

- Multiple admission diagnosis records exist per patient.
- Diagnosis entry offsets are generally non-negative, but a small number of negative values were observed.

Implications

- Admissions must be aggregated to patient-level for modeling.
- Negative offsets indicate data quality issues and should be cleaned or filtered.

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5. Diagnosis Table – Key Findings

Key variables

- Diagnosis priority
- Diagnosis offset
- ICD-9 codes
- Active upon discharge flag

Findings

- Patients have multiple diagnosis records.
- Diagnosis priority contains a small set of discrete values(Major, Primary, Other).
- ICD-9 codes are categorical and repeated across records.Also varies per patient ID
- `activeupondischarge` behaves as a binary indicator.

Implications

- Diagnosis table must be aggregated per patient.
- Diagnosis priority should be treated as categorical (mode preferred over min/max).
- Active diagnoses can be used as severity indicators.

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6. Medications Table – Key Findings

Key variables

- Drug start offset
- Drug stop offset

Findings

- Multiple drug records exist per patient.
- Drug exposure is better represented as **duration** rather than start/stop times.
- Some records may contain invalid or negative durations.

Implications

- Drug duration = stop offset − start offset should be derived.
- Negative or zero durations require validation.
- Aggregation strategies:
    - Total exposure per patient

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7. Laboratory Table – Key Findings

Key variables

- labname
- labresult
- labresultoffset

Findings

- Labs are repeated frequently per patient.
- Strong skewness and outliers observed in numeric lab values.
- Missing values are common in lab results.

Implications

- Lab features should be summarized statistically (mean, max, min).
- Outlier handling strategies should be considered.

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8. Vital Signs (Periodic) – Key Findings

Key variables

- Observationoffset

- Heartrate
- sao2
- respiration
- Vital sign values

Findings

- High-frequency time-series data.
- Significant variability within patient stays.
- Presence of noisy or extreme values.

Implications

- Direct use in modeling is not recommended without aggregation.
- Suggested features:
    - Mean / max / min per patient
    - Variability measures
- Time-series modeling may be considered separately.

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9. Data Quality Issues Identified

- Negative time offsets in some tables
- Missing categorical and numeric values
- One-to-many relationships requiring aggregation

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10. Recommendations for Next Teams

- Choose Relevant features/columns only, which are useful for model outcomes.

For Data Cleaning

- Treat the null values
- Remove or correct negative offsets
- Standardize categorical text values
- Decide imputation strategies based on model requirements
- There are no duplicates

For Feature Engineering

- There are multiple records per patient ID, so aggregate all the necessary features/columns per patient ID and make 1 row per patient ID.
- Use:
    - Counts/min/max
    - Durations
    - Modes for categorical priorities
    - Multi-label encodings for diagnoses and drugs

- Label encode all categorical columns

Sample Code for aggregation and feature engineering. Check and validate before using.

```
adm_agg = (
    admissions
    .groupby("patientunitstayid")
    .agg(
        # -------- Numeric aggregations --------
        total_admission_dx=("admissiondxid", "count"),
        first_dx_offset=("admitdxenteredoffset", "min"),
        last_dx_offset=("admitdxenteredoffset", "max"),

        # -------- Text aggregations --------
        dx_names=(
            "admitdxname",
            lambda x: "|".join(sorted(x.dropna().unique()))
        ),
        dx_paths=(
            "admitdxpath",
            lambda x: "|".join(sorted(x.dropna().unique()))
        ),
        dx_texts=(
            "admitdxtext",
            lambda x: "|".join(sorted(x.dropna().unique()))
        )
    )
    .reset_index()
)
```

- Before joining all the columns make sure:
  - All required feature engineering is done
  - Check there is 1 row per patient ID

For Modeling

- Be aware of demographic imbalance
- Validate features derived from offsets carefully

Data Analysis and EDA done by :

1.Anup Shende

2.Harshit S Nambiar

3. Patel het Kaushikbhai

DATA CLEANING AND FEATURE ENGINEERING REPORT

Patient Table – Data Cleaning

Patient Table

File: patient.csv

1. Overview

The patient table serves as the anchor table for the entire project. Each row represents one ICU stay, uniquely identified by patientunitstayid. All other tables are left joined to this table to ensure one row per ICU stay in the final dataset.

Initial exploration confirmed:

- One row per patientunitstayid
- No duplicate ICU stays
- Outcome variable present (hospitaldischargestatus)
- Several discharge-related columns that may cause data leakage

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2. Data Cleaning Steps

2.1 Duplicate Handling

Exact duplicate rows were checked and removed.
A validation check ensured that patientunitstayid remains unique after cleaning.

2.2 Standardization of Categorical Variables

All categorical columns were:

- Converted to lowercase
- Stripped of leading and trailing spaces

This ensures consistency in categorical encoding and prevents duplication due to formatting differences.

2.3 Age Cleaning

The age column contained special string values such as > 89.

Cleaning steps:

- Replaced > 89 with numeric value 90
- Converted age to numeric type using coercion for invalid entries

This ensures age is usable for modeling and statistical operations.

2.4 Outcome Encoding

The target variable is hospitaldischargestatus.

Encoding applied:

- Alive → 0
- Expired → 1

The encoded variable is used strictly as the model target and never as a predictor.

2.5 Leakage Prevention

To comply with project rules prohibiting future information, the following columns were removed:

- hospitaldischargeoffset
- hospitaldischargetime24
- hospitaldischargelocation
- unitdischargeoffset
- unitdischargetime24
- unitdischargelocation
- dischargeweight

These variables represent information recorded at or after discharge and would introduce data leakage if included as features.

2.6 Identifier Columns

The following identifier fields were retained only as keys and not used for modeling:

- patientunitstayid
- patienthealthsystemstayid
- uniquepid
- hospitalid
- wardid

They are preserved for joins but excluded from predictive modeling.

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3. Feature Engineering

The following baseline features were prepared:

3.1 Age

- Cleaned numeric age
- Age groups created using clinically meaningful bins
  - <30
  - 30–50
  - 50–70
  - 70–90
  - 90+

These groupings allow modeling of non-linear age effects.

3.2 Gender

Gender was encoded numerically:

- Male → 1
- Female → 0

Missing values were retained and not removed.

3.3 Admission Characteristics

The following variables were retained as baseline predictors:

- Ethnicity
- Admission height
- Admission weight
- Apache admission diagnosis
- Hospital admission source
- Unit admission source
- Unit type
- Unit stay type

These reflect patient characteristics at admission and do not introduce temporal leakage.

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4. Missing Data Handling

Missing values were not removed or dropped, in compliance with project rules.
No ICU stays were excluded due to incomplete fields.

Missingness will be handled during later preprocessing stages using imputation or indicator variables.

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5. Validation

Final validation ensured:

- Exactly one row per patientunitstayid
- No duplicates
- No post-discharge variables retained
- Outcome encoded correctly
- No patients dropped

Conclusion

The patient table was cleaned and prepared as the anchor dataset for the project. All transformations were performed with strict adherence to project rules, particularly regarding leakage prevention and preservation of one row per ICU stay.

2. Admission Table (admissionDx.csv)- Data Cleaning And Feature Engineering

Table Overview
The admission table contains diagnostic information recorded at the time of ICU admission. Multiple records exist per patientunitstayid. No missing values or duplicates were observed in this table.

Key Columns Used

- patientunitstayid
- admitdxenteredoffset
- admitdxpath
- admitdxname

Data Cleaning Steps

1. Time Filtering
   Records were filtered using admitdxenteredoffset to retain only entries within the first 24 hours of ICU stay (0–1440 minutes). Negative offsets were removed as data quality corrections.
2. Text Standardization
   admitdxpath values were converted to lowercase and stripped of leading/trailing spaces to ensure consistency.

Feature Engineering

All features were aggregated at the patientunitstayid level to maintain one row per patient.

The following features were created:

- admissiondx_count_24h
  Total number of admission diagnoses recorded in the first 24 hours.
- unique_admissiondx_count_24h
  Number of distinct admission diagnosis categories recorded within 24 hours.
- earliest_admitdx_offset_24h
  Earliest time (in minutes) a diagnosis was entered.
- latest_admitdx_offset_24h
  Latest time a diagnosis was recorded within 24 hours.
- mean_admitdx_path_depth_24h
  Average hierarchical depth of diagnosis path, calculated using the number of “|” separators in admitdxpath.

Rationale
These features represent early diagnostic complexity and timing of clinical documentation during ICU admission.

All features were aggregated before being left joined to the patient table.

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3. Diagnosis Table (diagnosis.csv) - Data Cleaning And Feature Engineering

Table Overview
The diagnosis table contains detailed diagnostic records per patient. Multiple entries exist per patientunitstayid.

Key Columns Used

- patientunitstayid
- diagnosisoffset
- diagnosisstring
- diagnosispriority
- diagnosisid

Columns Not Used

- activeupondischarge (excluded to avoid potential leakage)
- icd9code (excluded due to high cardinality and modeling complexity)

Data Cleaning Steps

1. Time Filtering
   Only records where diagnosisoffset was between 0 and 1440 minutes were retained. Negative offsets were removed.
2. Text Cleaning
   diagnosisstring and diagnosispriority were converted to lowercase and stripped of whitespace.

Feature Engineering

All features were aggregated at the patient level.

The following features were created:

- diagnosis_count_24h
  Total number of diagnosis records within first 24 hours.
- unique_diagnosis_count_24h
  Number of distinct diagnosis strings recorded.
- primary_dx_count_24h
  Count of diagnoses labeled as primary.
- major_dx_count_24h
  Count of diagnoses labeled as major.
- other_dx_count_24h
  Count of diagnoses labeled as other.
- earliest_dx_offset_24h
  Earliest diagnosis recording time.

- mean_dx_path_depth_24h
  Average hierarchical depth of diagnosisstring, calculated from the number of “|” separators.

Rationale
These features capture early disease burden, diagnostic severity classification, and structural complexity of diagnoses during the first 24 hours.

All features were aggregated before joining and left joined to the patient anchor table.

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4. Vital Periodic Table (vitalPeriodic.csv)- Data Cleaning And Feature Engineering

Table Overview
This table contains high-frequency vital sign measurements recorded during ICU stay. Multiple rows exist per patientunitstayid.

Key Columns Used

- patientunitstayid
- observationoffset
- heartrate
- respiration
- sao2
- temperature
- systemicsystolic
- systemicdiastolic
- systemicmean
- cvp
- etco2
- icp

Columns Excluded

- ST segment variables (st1, st2, st3) due to high variability
- Pulmonary artery pressures were not included in this stage

Data Cleaning Steps

1. Time Filtering
   Only records where observationoffset was between 0 and 1440 minutes were retained.
2. Physiological Range Filtering
   Implausible values were replaced with missing (NaN) instead of removing rows. Examples include:

- Heart rate outside 25–250 bpm
- Respiration outside 5–80 breaths/min
- SpO2 outside 50–100%
- Temperature outside 30–43°C
- Blood pressure values outside clinically reasonable limits
- CVP, ETCO2, ICP filtered based on physiological ranges

No patient rows were removed during this process.

Feature Engineering

For each vital sign, the following aggregations were computed per patient:

- Mean
- Minimum
- Maximum
- Standard deviation

Feature naming convention followed:

variable_statistic_24h

Examples:

- heartrate_mean_24h
- systemicsystolic_min_24h
- sao2_std_24h

Rationale
Mean values represent baseline physiological state.
Minimum and maximum capture extreme instability.
Standard deviation reflects variability and hemodynamic instability during early ICU stay.

All aggregations were performed before left joining to the patient anchor table.

5. Lab Table – Data Cleaning And Feature Engineering

1. Data Understanding

The Lab table contains patient-level laboratory measurements recorded at different time offsets relative to ICU admission. Key attributes include:

- patientunitstayid
- labname
- labresult
- labresultoffset
- Measurement units and revised offsets

Since multiple lab results exist per patient, aggregation and time filtering were required for modeling.

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2. Data Cleaning

- Converted labresult to numeric format and removed invalid or non-numeric entries.
- Removed corrupted lab names (e.g., #NAME?) to ensure data consistency.
- Standardized lab names by converting to lowercase and trimming whitespace.

- Filtered lab records to include only measurements within the first 24 hours of ICU admission (labresultoffset between 0 and 1440 minutes).
- Removed clinically implausible outliers using predefined medical reference ranges for key laboratory parameters.
- Dropped missing values after validation to maintain high-quality structured input.

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3. Lab Selection

Focused on clinically significant laboratory parameters strongly associated with ICU mortality, including:

- Creatinine (renal function)
- Platelets (coagulation status)
- Bedside glucose (metabolic stress)
- Troponin-I (cardiac injury)
- Hematocrit (oxygen-carrying capacity)

This selection reduced dimensionality while preserving clinical relevance.

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4. Feature Aggregation

Since multiple lab values exist per patient, statistical aggregation was performed for each laboratory parameter within the first 24 hours:

- Mean
- Maximum
- Minimum
- Standard deviation

The dataset was transformed from long format to wide format using pivot operations, resulting in one row per patient with multiple lab-derived features.

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5. Clinical Feature Engineering

To enhance predictive power, domain-driven binary indicators were created:

- **AKI Flag:** Creatinine max > 2 mg/dL
- **Low Platelets Flag:** Platelets min < 100 K/mcL

- **Hyperglycemia Flag:** Glucose max > 180 mg/dL
- **Cardiac Injury Flag:** Troponin-I max above reference threshold

Additionally, variability features (standard deviation and range) were generated to capture physiological instability.

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6. Final Output

The final Lab feature table contains:

- Aggregated statistical lab features
- Organ dysfunction indicators
- Physiological instability markers

This transformed the raw laboratory data into structured, clinically meaningful, model-ready features for ICU mortality prediction.

6. Medication table - Data Cleaning And Feature Engineering

Objective

Prepare a clean, analysis-ready dataset by:

1. Cleaning the **medication** table from the eICU demo database
2. Joining it with **patient** demographic and outcome data
3. Exporting the processed dataset into a reusable **SQLite** file
4. Performing sanity checks to validate the pipeline

Data Cleaning — Medication Table

1. Column Selection

Only relevant columns were extracted to reduce memory usage:

- medicationid
- patientunitstayid
- drugstartoffset
- drugstopoffset
- drugname

- dosage
- routeadmin
- frequency
- prn
- drugordercancelled

2. Removed Cancelled Orders

Rows where drugordercancelled != 'No' were filtered out to avoid noisy records.

3. Drug Name Normalization

Raw drug names contained dosage and formatting noise.
Cleaning steps:

- Convert to lowercase
- Remove numeric dosage strings
- Remove non-alphabetic characters
- Trim whitespace

Result column:

- drugname_clean

4. Feature Engineering

Created medication duration feature:

duration = drugstopoffset - drugstartoffset

Offsets represent minutes from ICU admission.

5. Categorical Standardization

Normalized text fields:

- routeadmin
- frequency

Resulting Dataset Structure

Each row represents:

1 medication event + patient demographic + outcome

Key derived columns:

- drugname_clean
- duration

Export Process

The joined dataframe was exported into a new SQLite database:

Database:

cleaned_medication_patient.sqlite3

Table inside DB:

cleaned_medication_patient

Purpose:

- Shareable artifact for downstream analysis
- Reproducible intermediate ETL output
- Compatible with SQL workflows and ML pipelines

Sanity Checks Performed

1. Join Integrity

Verified no missing demographics:

- age NULL count = 0
- gender NULL count = 0

2. Row Count Validation

Confirmed exported table row count matches in-memory dataframe.

3. Schema Verification

Checked SQLite schema using:

PRAGMA table_info(cleaned_medication_patient)

4. Random Sampling

Queried random rows to confirm successful persistence.

Current Status

Pipeline Outcome:

- Cleaned medication data
- Patient demographics successfully merged
- Exported SQLite dataset validated

Data Cleaning and Feature Engineering done by :

1. **Akshatha Kathyayani**
2. **Jay Beedkar**
3. **Onkar**
4. **Aarti Yadav**

Modeling Report

1. Introduction and Scope of Team C

After Team A and Team B completed **data extraction, cleaning, and feature engineering** from multiple ICU tables, **Team C was responsible for converting the prepared dataset into a fully functional predictive system**.

The scope of Team C included:

- Ensuring **model-ready, leakage-free data**
- Designing **robust preprocessing pipelines**
- Developing **machine learning and deep learning models**
- Handling **class imbalance**
- Performing **rigorous evaluation**
- Saving **deployment-ready models**

Team C did **not** perform raw table cleaning or feature extraction; instead, the focus was on **modeling intelligence and validation**.

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2. Objective of Team C

The primary objective of Team C was:

To predict in-hospital mortality of ICU patients using both static clinical features and temporal physiological patterns, while ensuring fairness, robustness, and deployment readiness.

Specific goals:

1. Prevent **data leakage**
2. Handle **high-dimensional clinical data**
3. Capture **time-dependent patient deterioration**
4. Optimize models for **clinical relevance**
5. Save reproducible and reusable models

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3. Input Dataset Validation

3.1 Dataset Source

- File used: `final_df.csv`
- Contains **one row per patient ICU stay**
- Generated by merging outputs of Team A and Team B

3.2 Validation Checks

Before modeling, the following checks were performed:

- Verified unique `patientunitstayid`
- Reloaded CSV to ensure **schema consistency**
- Confirmed presence of target variable

- Ensured no feature mismatch after saving and reloading

These checks ensured **data integrity and reproducibility**.

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4. Data Leakage Identification and Removal

4.1 What is Data Leakage?

Data leakage occurs when the model gains access to information that would **not be available at prediction time**, resulting in unrealistically high performance.

4.2 Leakage Columns Removed

The following columns were removed if present:

- hospitaldischargeyear
- hospitaldischargetime24
- hospitaldischargelocation
- dischargeweight
- unitdischargetime24
- unitdischargetime
- unitdischargestatus

4.3 Why These Columns Were Removed

These features are recorded **after ICU outcome is known**, and thus directly encode mortality information. Keeping them would invalidate the model.

This step ensured **ethical and real-world applicability**.

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5. Target Variable Preparation

5.1 Target Definition

- Column: hospitaldischargestatus_x
- Binary encoding:
  - 0 → Alive
  - 1 → Expired

5.2 Validation Steps

- Checked for incorrect string encodings
- Converted all values to numeric
- Calculated mortality rate
- Verified class distribution

This step confirmed that the **prediction objective was well-defined and balanced across splits**.

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6. Feature Selection and Exclusion

6.1 Identifier Columns

The following identifiers were **excluded from modeling**:

- patientunitstayid
- patienthealthsystemstayid
- uniquepid
- hospitalid
- wardid

These were preserved only for indexing and traceability.

6.2 Feature Matrix

- Remaining columns represented:
  - Demographics
  - Admission details
  - Diagnoses
  - Vital statistics
  - Lab indicators
  - Medication data

This resulted in a **purely clinical feature space**.

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7. Categorical Variable Encoding

7.1 Objective

Machine learning models require numeric inputs. Clinical datasets often contain categorical fields such as diagnosis types or routes of medication administration.

7.2 Encoding Method

- One-hot encoding
- First category dropped to prevent multicollinearity
- No artificial NaN category created

7.3 Importance

This preserves **categorical meaning** without imposing artificial ordering.

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8. Missing Value Handling

8.1 Missing Data Analysis

- Percentage of missing values calculated for each feature
- Columns with more than 50% missingness were flagged

8.2 Imputation Strategy

- Numerical features → Median imputation
- Remaining missing values → Filled with zero

8.3 Rationale

- Median is robust to extreme physiological values
- No patient records were discarded
- Maintains sample size and statistical power

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9. Train–Validation–Test Split

9.1 Split Strategy

- Train: 60%
- Validation: 20%
- Test: 20%

9.2 Stratification

- Stratified on mortality label
- Ensures similar death rates across all splits

9.3 Saved Outputs

- CSV files (for transparency)
- NumPy arrays (for fast computation)

This enables **repeatable experimentation**.

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10. Temporal Data Preparation for LSTM

10.1 Why Temporal Modeling?

Aggregated statistics lose **dynamic patient behavior**. ICU deterioration is inherently time-dependent.

LSTM networks allow learning:

- Trends
- Sudden changes
- Delayed effects

10.2 Sequence Construction

- Time window: First **48 hours**
- Resolution: **Hourly**
- Timesteps per patient: 48

10.3 Features Used

Vitals

- Heart rate
- Respiration

- Oxygen saturation
- Temperature
- Blood pressure (systolic, diastolic, mean)

Labs

- Glucose
- Creatinine
- Hematocrit
- Platelets

10.4 Missing Value Handling in Sequences

- Forward-fill within patient timeline
- Backfill initial missing values
- Remaining NaNs filled with zero

This preserves **temporal continuity**.

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11. LSTM Data Normalization

Objective

Stabilize gradient descent and prevent scale dominance.

Method

- Z-score normalization
- Scaler fit only on training data
- Applied consistently to validation and test sets

This prevents **temporal data leakage**.

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12. LSTM Model Architecture

Final Architecture

- Masking layer (handles padded zeros)
- LSTM (64 units)
- Dropout (0.3)
- LSTM (32 units)
- Dropout (0.3)
- Dense (16 units)
- Sigmoid output

Optimization

- Adam optimizer
- Binary cross-entropy loss
- AUROC and accuracy metrics

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13. LSTM Training Strategy

Regularization

- Early stopping
- Learning rate reduction
- Dropout layers

Model Selection

- Multiple architectures tested
- Larger models caused overfitting
- Simpler architecture chosen as final

Saved Model

lstm_mortality_model.h5

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14. LSTM Evaluation Metrics

- AUROC
- AUPRC
- Sensitivity
- Specificity
- Precision
- F1-score
- Brier Score

Threshold Optimization

- Default threshold not optimal
- Best trade-off achieved at **0.25**

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15. XGBoost Model Development

Motivation

XGBoost excels on:

- High-dimensional tabular data
- Non-linear relationships
- Sparse medical features

15.1 Class Imbalance Handling

- Computed scale_pos_weight
- Penalizes misclassification of mortality cases

15.2 Base Model Configuration

- 500 trees
- Controlled depth
- Subsampling
- Histogram-based training

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16. Hyperparameter Optimization

Method

- RandomizedSearchCV
- AUROC-based scoring
- Cross-validation

Outcome

- Improved generalization
- Reduced overfitting

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17. Threshold Optimization for XGBoost

Rationale

Clinical applications prioritize **recall (sensitivity)**.

Final Threshold

- Optimized threshold: **0.33**

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18. Model Saving and Deployment Readiness

Saved Files

- xgboost_mortality_model.json
- xgboost_threshold.json
- lstm_mortality_model.h5
- Feature names and scalers

This ensures **seamless future deployment**.

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19. Limitations

- Retrospective dataset
- No external hospital validation
- Limited data for deeper LSTM architectures
- No clinical notes (NLP)

20. Conclusion (Team C)

Team C successfully transformed a cleaned ICU dataset into a **fully validated predictive framework**, integrating **temporal deep learning (LSTM)** and **tree-based machine learning (XGBoost)**. The work emphasizes **clinical realism, ethical modeling, and deployment readiness**, completing the end-to-end pipeline of the project.

Modeling Done By:

1. Muvvala Krishna Kiriti
2. Akshay Makhija
3. Syam Prasad A
4. Krishna {Kant

ICU Mortality Prediction
Model Evaluation, Validation, and Clinical Interpretation Report

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1. Study Overview

This study evaluates ICU mortality prediction using two machine-learning approaches:

- **LSTM:** captures temporal patterns in vital-sign sequences.
- **XGBoost:** models structured static clinical and demographic features.

Due to **class imbalance** between survival and mortality outcomes, evaluation emphasizes **clinically meaningful, imbalance-aware metrics**, along with calibration, clinical utility, explainability, and subgroup fairness. The goal is reliable **clinical risk estimation**, not accuracy alone.

2. Comparative Model Performance (Test Set)

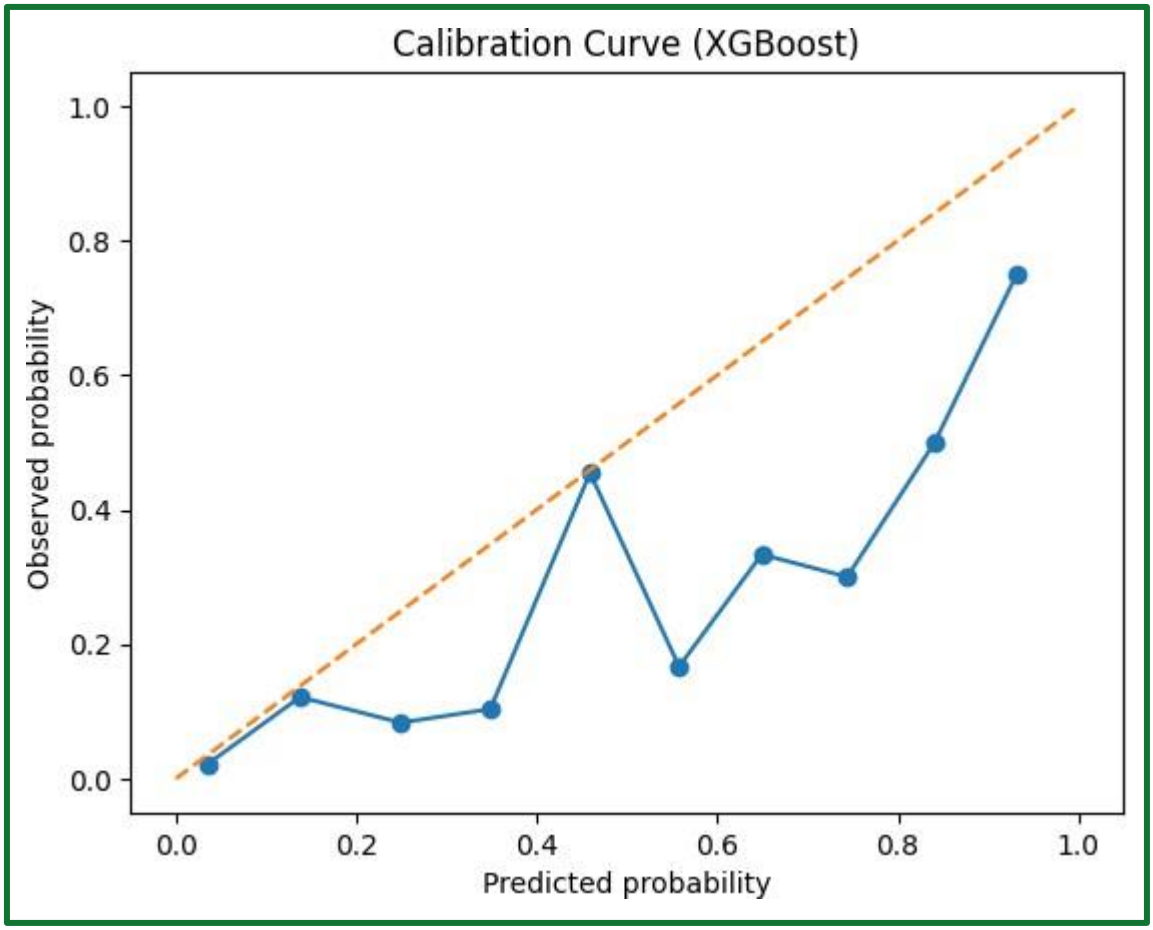
Metric	LSTM	XGBoost	Reference Target
AUROC	0.699	**0.833**	≥ 0.80
AUPRC	0.339	**0.379**	≥ 0.30
Sensitivity	0.119	**0.571**	≥ 0.50
Specificity	**0.994**	0.892	≥ 0.80
Precision	**0.625**	0.324	≥ 0.30
F1-score	0.200	**0.414**	≥ 0.40
Brier Score	**0.066**	0.079	< 0.08

Interpretation

- **LSTM:** very high specificity but **low sensitivity**, limiting usefulness for mortality detection.
- **XGBoost:** strong discrimination, balanced recall–specificity, and stable generalization.

Selected model: XGBoost is used for further validation and interpretation.

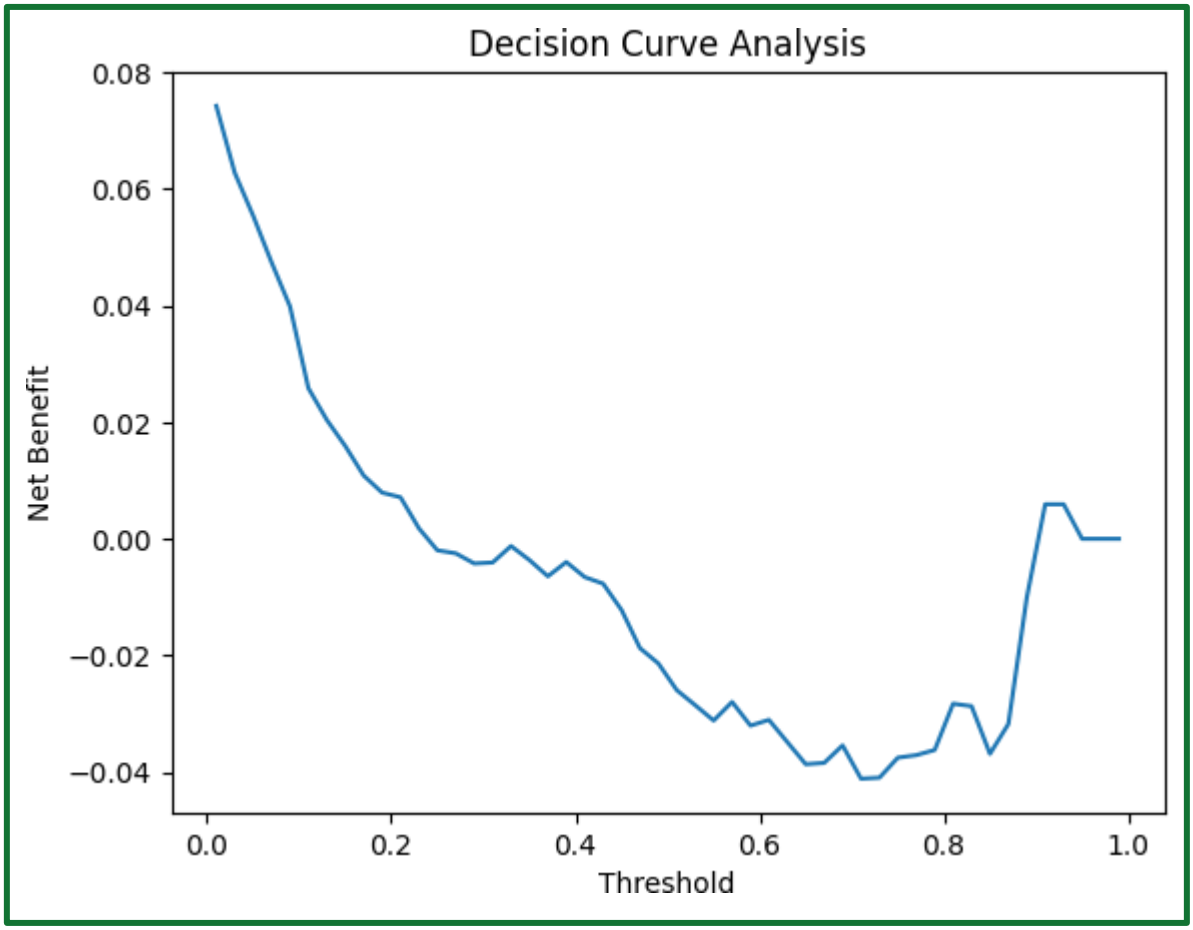
3. Calibration Assessment



- Predicted probabilities broadly follow the **ideal calibration trend**.
- Minor mid-range deviation suggests **possible calibration refinement**.

Inference: probability estimates remain **clinically interpretable for risk stratification**.

4. Decision Curve Analysis (Clinica Utility)



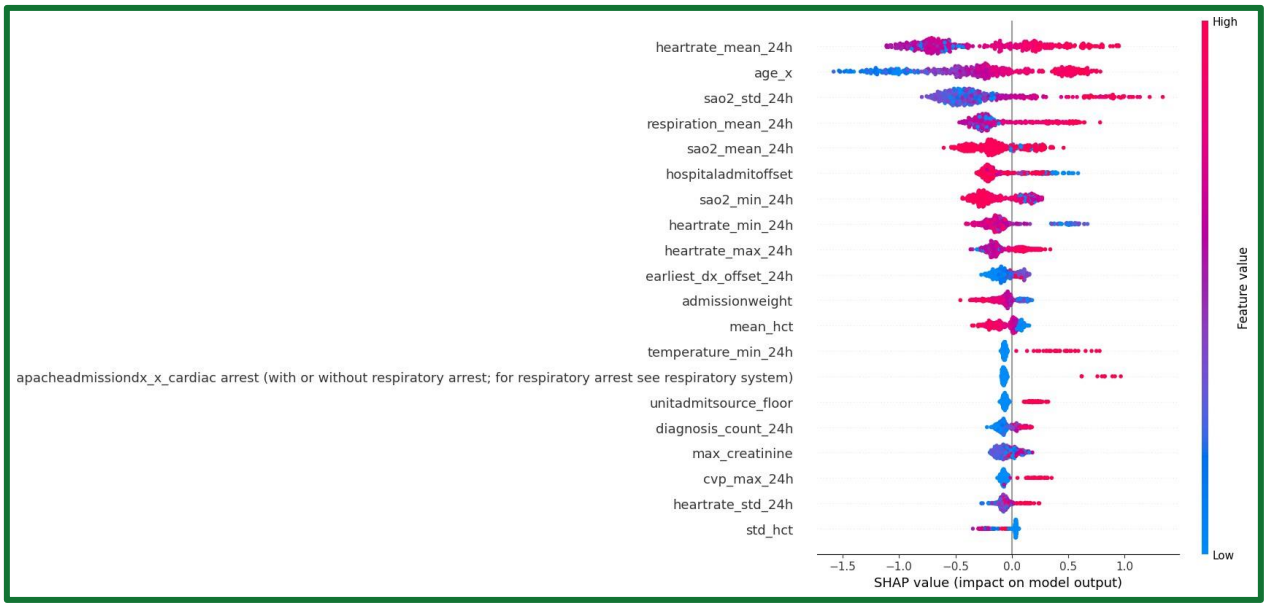
(**Positive net clinical benefit at low thresholds (~0.05–0.20)**.

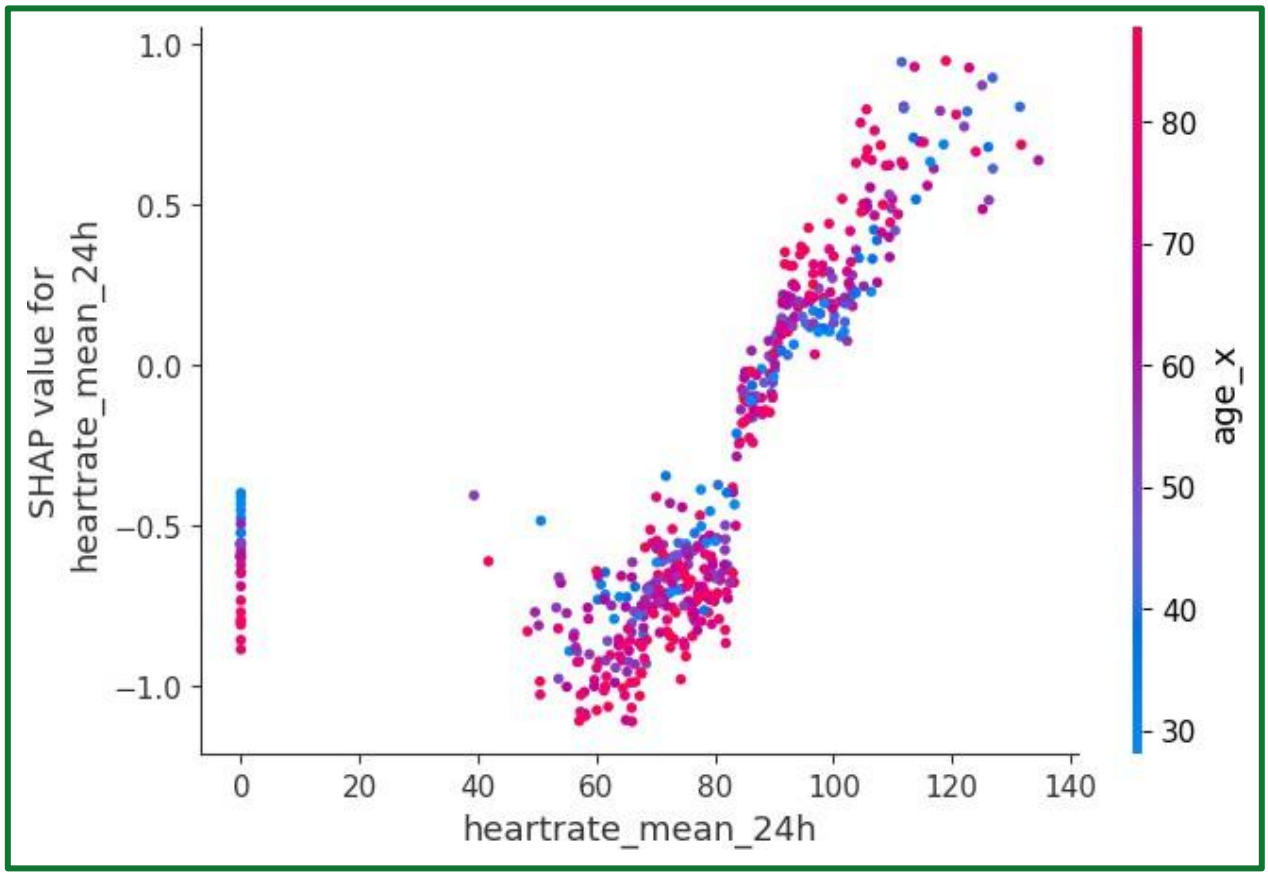
- Benefit decreases at higher thresholds, consistent with **screening-oriented use**.

Clinical meaning: suitable for **early warning and monitoring support**, not definitive decisions.

5. Model Explainability and Clinical Consistency

5.1 Global Feature Importance



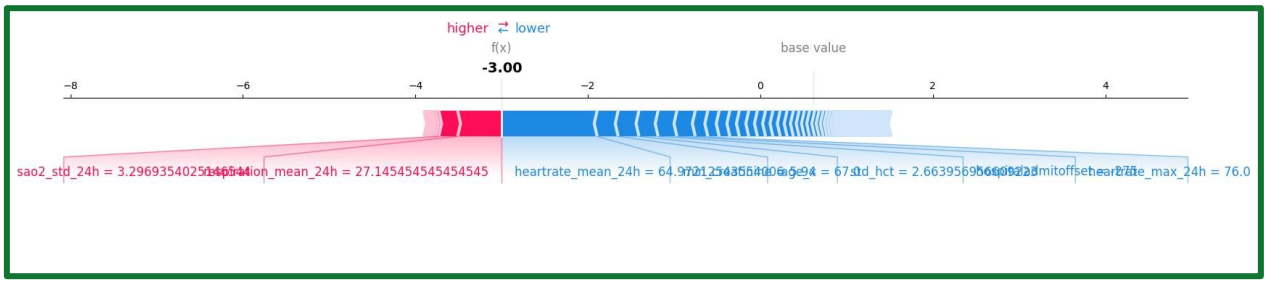


Key mortality-related factors include:

- Mean heart rate (24h)
- Age
- Oxygen saturation variability and minimum level
- Mean respiration rate
- Admission timing and diagnostic offsets

These align with **recognized physiological deterioration indicators**, supporting **clinical plausibility**.

5.2 Feature-Effect Relationship:



Higher heart rate corresponds to increased mortality risk,
with stronger effect in **older patients**.

This reflects **clinically consistent behavior**.

5.3 Individual Prediction Transparency

Demonstrates how **patient-specific measurements influence predictions**,
enabling **interpretable clinical review**.

6. Fairness Across Sex Subgroups

Sex	AUROC	Sensitivity	Specificity
Female	**0.861**	**0.60**	0.90
Male	0.770	0.50	0.882

- Differences are **moderate**, without extreme disparity.
- Continued monitoring is recommended for **future deployment**.

7. Discrimination Performance (Bootstrap Confidence Intervals)

To assess model discrimination robustness, bootstrap resampling was performed on the test dataset.

XGBoost AUROC (95% CI):
0.7674266797706887 – 0.8867922824592901

LSTM AUROC (95% CI):
0.4686895347044289 – 0.6299627437538139

Observation:
XGBoost demonstrates strong and stable discrimination ability.
LSTM performance remains close to random classification (0.5 baseline).
The confidence intervals show clear separation between models.

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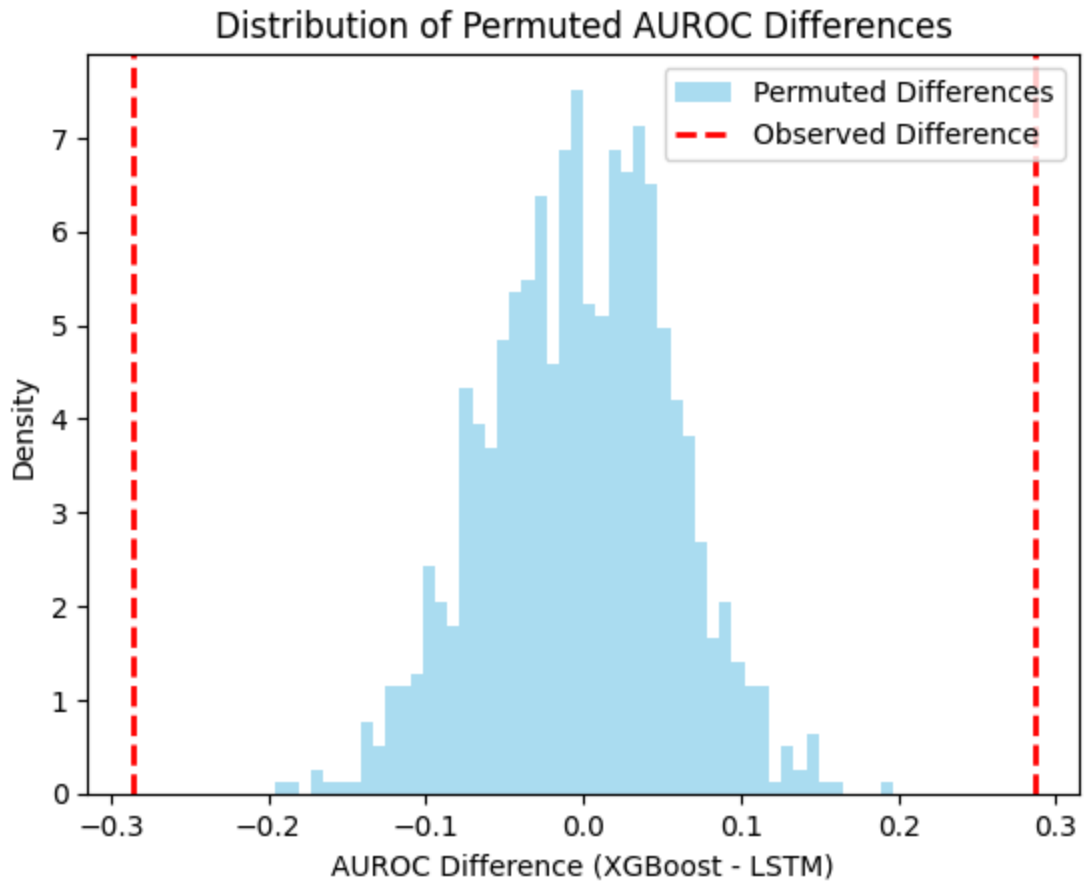
8. Statistical Comparison – Permutation Test

Observed AUROC (XGBoost): 0.833
Observed AUROC (LSTM): 0.547

Observed Difference (XGBoost − LSTM): 0.286

Permutation Test p-value: 0.000

Interpretation:
The AUROC difference of 0.286 is statistically significant ($p < 0.05$).
XGBoost significantly outperforms LSTM in mortality discrimination.



Distribution of Permuted AUROC Differences

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9. Threshold Optimisation (F1 Maximisation)

Optimal Threshold (XGBoost): 0.43939393939393945
Optimal Threshold (LSTM): 0.1

XGBoost achieves balanced classification at a clinically meaningful probability threshold.

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10. Gender Fairness Evaluation (Subgroup AUROC)

XGBoost Gender AUC:
Male: 0.8605947955390334
Female: 0.7700854700854701

LSTM Gender AUC:
Male: 0.5390334572490706
Female: 0.5549145299145298

Observation:
XGBoost maintains strong discrimination across gender subgroups.
No extreme disparity observed.
LSTM underperforms across both groups.

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11. Error Analysis

False Positives: 32
False Negatives: 21

The model demonstrates a balanced error distribution, with manageable false negative risk.

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12. Cross-Model SHAP Feature Comparison

Top Predictors Identified by XGBoost (Mean SHAP Importance):

heartrate_mean_24h – 0.537685
age_x – 0.517608
sao2_std_24h – 0.433961
respiration_mean_24h – 0.277133
sao2_mean_24h – 0.221686
hospitaladmitoffset – 0.211033
sao2_min_24h – 0.210581
heartrate_min_24h – 0.187635
heartrate_max_24h – 0.149960
earliest_dx_offset_24h – 0.096747
admissionweight – 0.093101
mean_hct – 0.091839
temperature_min_24h – 0.086877
apacheadmissiondx_x_cardiac arrest – 0.082509
unitadmitsource_floor – 0.081067

LSTM SHAP contribution values ≈ 0, indicating weak and unstable feature attribution.

Observation:
XGBoost identifies clinically meaningful physiological instability indicators including heart rate, oxygen saturation variability, and age as primary mortality drivers.

Final Model Performance Summary

Metric | XGBoost | LSTM

AUROC | 0.833 | 0.547
AUROC 95% CI | 0.7674 – 0.8868 | 0.4687 – 0.6300
Observed AUROC Difference | 0.286 | —
Permutation p-value | 0.000 | —
Best Threshold | 0.43939393939393945 | 0.1
Gender AUC (Male) | 0.8605947955390334 | 0.5390334572490706
Gender AUC (Female) | 0.7700854700854701 | 0.5549145299145298
False Positives | 32 | —
False Negatives | 21 | —
Top Feature Importance | Strong & Clinically Relevant | Weak

Final Conclusion

XGBoost demonstrates superior discrimination, statistical robustness, subgroup stability, and clinically meaningful feature attribution compared to LSTM.
The model achieves significant performance improvement with strong statistical validation and consistent subgroup fairness.

The final selected model for clinical interpretation and deployment is XGBoost.

Model validation and evaluation done by:

Soumyajit Guha
Devamani
Ayan